Selection on Observables II: Estimation

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Where we left off

Conditional ignorability (CI):

$$\{Y_{1i}, Y_{0i}\} \perp\!\!\!\perp D_i \mid X_i$$

Under CI (+ common support), the ATE is identified:

$$\mathbb{E}[Y_1 - Y_0] = \mathbb{E}_X[\,\mathbb{E}[Y \mid D = 1, X] - \mathbb{E}[Y \mid D = 0, X]\,]$$

In words: an <u>experiment inside each stratum of X</u>, then average over X.

Understanding and arguing about CI is the hard and important thing.

Today: how to actually do the adjustment on the chosen X

Roadmap

A menu of ways to turn CI into an estimate:

- **Subclassification** (last time)
- **Matching** (a flexible version of subclassification)
- **Propensity score** methods (matching, then IPW)
- **Balancing weights** (skip the propensity score; balance directly)
- **Regression** (via a “regression imputation” lens)

Under CI, these are all estimating the same thing. They differ only in *how they approximate conditioning on X*.

Recap: Subclassification

From SOO I:

- Partition units by X into strata $x_1, x_2, \ldots$
- Within each stratum, treated and control are exchangeable (CI).
- Compute DIM within each stratum; weight up to the population:

$$\widehat{\tau} = \sum_{x} \left[\bar{Y}_1(x) - \bar{Y}_0(x) \right] \widehat{\Pr}(X = x)$$

Works great when X is low-dimensional and discrete.

Problematic with too many variables, continuous variables: empty and under-populated strata.

Matching: subclassification with 1-unit "strata"

- For each treated unit i, find the closest control unit j in X.
- The pair (i, j) is like a mini, approximate stratum.
- ATT estimator:

$$\widehat{\tau}_{ATT} = \frac{1}{n_1} \sum_{i:D_i=1} \left(Y_i - Y_{j(i)} \right)$$

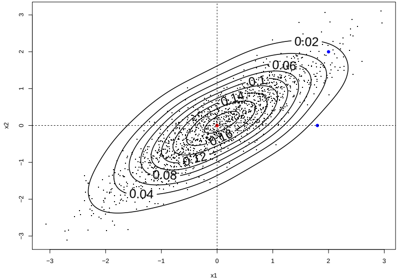
A matched sample is just a flexible subclassification. Same idea; the strata move with the data.

What is “closest”?

With a single X: obvious — smallest $|X_i - X_j|$.

With several X: need a distance metric.

- **Euclidean**: $\|X_i - X_j\|$ — scale-sensitive.
- **Mahalanobis**: $\sqrt{(X_i - X_j)'\widehat{\Sigma}_X^{-1}(X_i - X_j)}$ — scale-invariant, accounts for correlation.



The limits of matching

Problem: unlike true stratification, the treated and control getting compared aren't exactly the same on X_i.

- There is a *discrepancy*, $\|X_i - X_j\|$, between treated and control units in a pair $\rightarrow$ difference in $Y_i(1)$ and $Y_j(0)$. This bias does not vanish at the usual $\sqrt{n}$ rate (Abadie & Imbens, 2006).
- As $\dim(X)$ grows, the distance to even your nearest neighbor rapidly increases ("curse-of-dimensionality").

One solution: **bias adjustment** (Abadie & Imbens, 2011).

- Regress $Y \sim X$ among controls; call the fit $\widehat{\mu}_0(X)$.
- Use $\widehat{\mu}_0(X_i) - \widehat{\mu}_0(X_{j(i)})$ to estimate the $Y(0)$ gap implied by the X-discrepancy, and subtract it from each matched pair.

The propensity score

Another solution: Find just one dimension that really counts.

The propensity score is the probability of treatment given X:

$$\pi(X) \ = \ \Pr(D = 1 \mid X)$$

Rosenbaum & Rubin (1983): if CI holds given X, it also holds given $\pi(X)$.

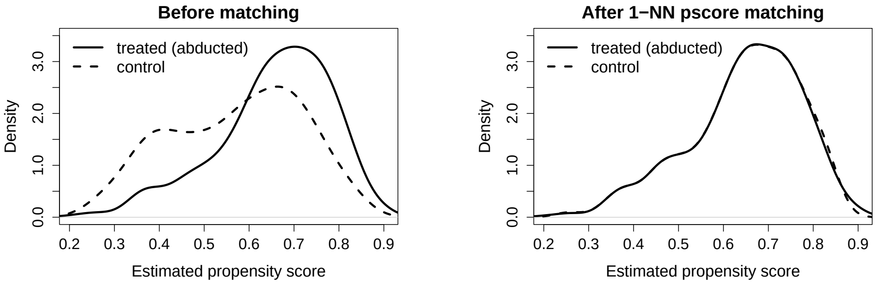
$$\{Y_1, Y_0\} \perp\!\!\!\perp D \mid X \quad \Longrightarrow \quad \{Y_1, Y_0\} \perp\!\!\!\perp D \mid \pi(X)$$

Instead of matching in high-dimensional X, just a scalar!

Propensity score matching, on Blattman & Annan example

Recipe:

1. Fit $\widehat{\pi}(X)$ (e.g., logistic regression of D on X).
2. For each treated i, find the control j with the closest $\widehat{\pi}(X_j)$.
3. Compute DIM on the matched sample.



Caveats: Extreme π may need to be trimmed; depends on how well we estimate $\pi(X)$

From matching to weighting

Is there a better way to use the pscore without the wasted sample and discrepancy from matching?

Intuition: rare treated units (low π) represent many similar untreated units. So count them more.

IPW weights:

$$w_i = \frac{D_i}{\widehat{\pi}(X_i)} + \frac{1 - D_i}{1 - \widehat{\pi}(X_i)}$$

Treated units with low π: upweighted (they're rare).
Control units with high π: upweighted (also rare).

Weights: What we want vs what we have

Under CI, the ATE is an average of within-X effects over the overall X-distribution:

$$\mathbb{E}[\tau] = \int \underbrace{\{\mathbb{E}[Y \mid D = 1, X] - \mathbb{E}[Y \mid D = 0, X]\}}_{\text{within-}X\text{ effect}} \; p(X)\, dx$$

But the raw DIM averages each arm over the wrong distribution:

$$\text{DIM} = \int \mathbb{E}[Y \mid D{=}1, X]\, p(X \mid D{=}1)\, dx \; - \; \int \mathbb{E}[Y \mid D{=}0, X]\, p(X \mid D{=}0)\, dx$$

Treated units live in $p(X \mid D{=}1)$; controls in $p(X \mid D{=}0)$. We want both to live in $p(X)$.

Weights can do this...

The weights that close the gap

To turn $p(X \mid D{=}1)$ into $p(X)$, multiply by $\frac{p(X)}{p(X \mid D{=}1)}$. Same idea for controls.

In either arm, such a weight has the form:

$$w_i \;=\; \frac{p(X_i)}{p(X_i \mid D_i)}$$

By Bayes' rule:

$$\frac{p(X_i)}{p(X_i \mid D_i)} \;=\; \frac{p(D_i)}{p(D_i \mid X_i)}$$

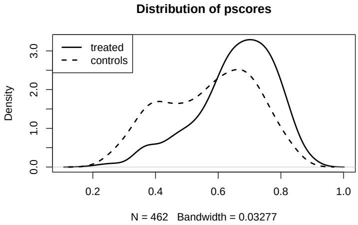
So the weight for each unit is $\frac{p(D_i)}{p(D_i \mid X_i)}$ — the **stabilized IPW weights**:

$$w_i^{\text{stab}} \;=\; \frac{D_i \; \Pr(D{=}1)}{\widehat{\pi}(X_i)} \;+\; \frac{(1 - D_i) \; \Pr(D{=}0)}{1 - \widehat{\pi}(X_i)}$$

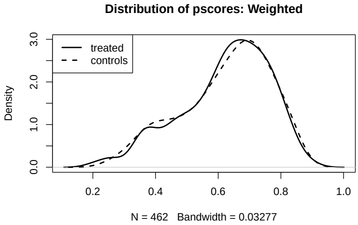
Drop the $p(D_i)$ in numerator and you'd have the plain IPW above ($1/\widehat{\pi}(X_i)$ or $1/(1 - \widehat{\pi}(X_i))$), which has higher variance.

IPW in pictures

Before weighting, treated and control differ in X:



After IPW, the control distribution is reshaped to match the treated:



IPW caveats

What can go wrong:

- **Extreme weights/ poor overlap**: $\widehat{\pi}(X) \approx 0$ or $1 \Rightarrow$ enormous weights $\Rightarrow$ huge variance, signal of non-overlap (extrapolation)
- **Misspecification.** $\widehat{\pi}(X)$ wrong $\Rightarrow$ weights wrong $\Rightarrow$ bias.
- Trimming is sometimes used to address extreme weights...changes the estimate, can be arbitrary.

Wait — why estimate $\pi(X)$ at all?

If you really just want get treated and control onto the same distribution in X, why not just weight to make it so?

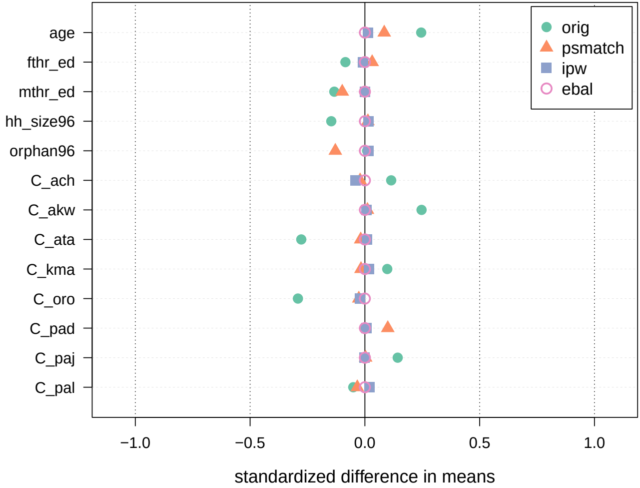
Choose weights w_i to satisfy, e.g., **mean balance on** X:

$$\sum_{i:D_i=1} w_i X_i = \sum_{i:D_i=0} w_i X_i$$

...subject to a regularizer (keep weights as close to uniform as possible).

E.g. maximum entropy balancing (Hainmueller, 2012) minimizes $\sum_i w_i log(w_i)$ while achieving the target balance conditions.

Balance on Blattman & Annan



Standardized mean differences, 13 covariates. “orig” = no adjustment.

How much balance?

Mean balance on X is not full distributional balance.

You can ask for more:

- higher moments of X,
- interactions,
- kernels / non-parametric balance (e.g. kernel balancing).

Tradeoff: you only ensure balance on the things you target, but you get perfect balance on them in the sample, don't need correct pscore.

Regression — start with imputation

What's actually missing? For each unit i with covariates X_i:

- If $D_i = 1$, we observe $Y_i(1)$; we don't know $Y_i(0)$.
- If $D_i = 0$, we observe $Y_i(0)$; we don't know $Y_i(1)$.

Idea: impute the missing potential outcome using $\mu_d(x) \equiv \mathbb{E}[Y(d) \mid X = x]$:

- Fit $\widehat{\mu}_1(x)$ on treated units (estimates $\mathbb{E}[Y(1) \mid X{=}x]$).
- Fit $\widehat{\mu}_0(x)$ on control units (estimates $\mathbb{E}[Y(0) \mid X{=}x]$).
- For each i, predict both: $\widehat{Y}_i(1) = \widehat{\mu}_1(X_i)$, $\widehat{Y}_i(0) = \widehat{\mu}_0(X_i)$.

Average the individual-level differences:

$$\widehat{\tau} = \frac{1}{n} \sum_i \left[\widehat{\mu}_1(X_i) - \widehat{\mu}_0(X_i)\right]$$

This is **regression imputation**, a.k.a. ***g*-computation**. As natural a use of regression as there is.

Linear models $\Rightarrow$ Lin's estimator

Suppose $\widehat{\mu}_1$ and $\widehat{\mu}_0$ are linear:

$$\widehat{\mu}_d(x) = \widehat{\alpha}_d + \widehat{\beta}_d' x, \quad d \in \{0, 1\}$$

Fitting separate linear models on treated and control and then averaging predicted differences is exactly the interacted OLS regression

$$Y = \alpha + \tau D + \beta' X + \gamma'(X \cdot D) + \varepsilon$$

evaluated at the overall mean of X.

This is **Lin's (2013) estimator** — which you already saw in experiments. The same thing, just now motivated by "impute each unit's missing PO."

What about pooled OLS?

The usual regression $Y = \alpha + \tau D + \beta' X + \varepsilon$ (no interaction):

- Assumes the same slope in X for treated and control.
- Equivalent to fitting a pooled $\widehat{\mu}(x)$ and taking $\widehat{\tau}$ from the D coefficient.
- When treatment effects vary with X: $\widehat{\tau}$ is a weighted average of unit-level effects, with weights that tilt toward units in the region of overlap — not necessarily the ATE or ATT.

Fix: either use the interacted (Lin) version, or commit to the assumption that slopes are equal.

All roads lead to the same place

Under CI, every method on the menu is trying to estimate

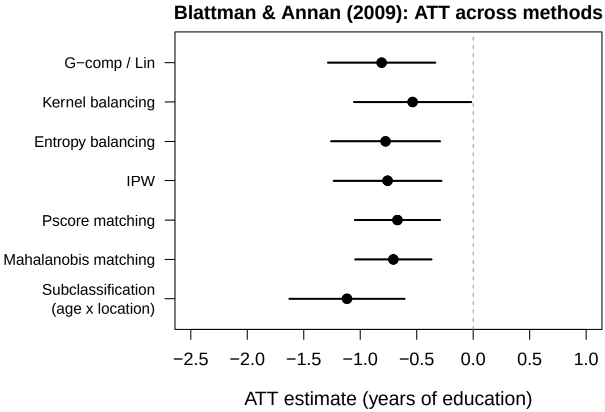
$$\mathbb{E}_{X\sim F}[\,\mu_1(X) - \mu_0(X)\,]$$

- F is a choice of estimand: $F = p(X)$ gives the ATE; $p(X \mid D{=}1)$ gives the ATT; $p(X \mid D{=}0)$ gives the ATC.
- the "adjusting for X" is borne out by using the same F for both treated and controls

They achieve this by different strategies:

Method	**How it conditions on X**
Subclassification	coarse strata of X
Matching	nearest-neighbor "strata" in X
Pscore matching	match on $\widehat{\pi}(X)$
IPW	reweight by $1/\widehat{\pi}(X)$
Balancing weights	solve for weights balancing X
Regression imputation	model $\mu_d(X)$ and impute

Blattman & Annan: ATT across methods



Covariates: age, fthr_ed, mthr_ed, hh_size96, orphan96, and 8 location dummies.

If everything agrees, great; if not, diagnose.

Beyond: combining pscore and outcome model

So far: estimate either $\widehat{\pi}(X)$ (weighting) <u>or</u> $\widehat{\mu}_d(X)$ (regression).

Doubly robust methods combine both:

- **AIPW**: IPW estimator + regression adjustment using $\widehat{\mu}_d(X)$.
- **TMLE**: iteratively update $\widehat{\mu}_d$ using $\widehat{\pi}$.
- **Double ML (DML)**: plug flexible (ML-based) $\widehat{\pi}$ and $\widehat{\mu}_d$ into AIPW with cross-fitting.

Consistent if *either* $\widehat{\pi}$ or $\widehat{\mu}_d$ is correctly specified. Good inference with flexible (ML) fits if both converge fast enough.

All still need CI + common support. Flexibility buys you less bias from model form, not from missing confounders.

Takeaways

- **CI does the identifying work.** Estimation is a menu of ways to operationalize "experiment within strata of X."
- **Subclassification** breaks in high-d; **matching** is the flexible version; the **propensity score** is a dimension reducer.
- **IPW** reweights rather than matches; **balancing weights** skip the $\pi(X)$ step and enforce balance directly.
- **Regression** is natural when framed as imputation: model $\mu_d(X)$, predict each unit's missing PO (or both), average over intended group.
- **Reasonable practice**: In a given setting, you might take one estimator as central, but good to show range of answers you get under whichever approaches are appropriate.